SmartAg Ops Advisor

Epics and user stories · based on the Master Document (25 Oct 2026)

Epic	Goal	Stories
E1 Accounts, Roles & Access Control	Safe sign-in, role-based access, and verified truck operators.	E1-S1 Sign up and sign in with role-based access E1-S2 Admin OTP sign-in and password recovery E1-S3 Verified truck-operator profile and documents
E2 Crop Stock & Market Price Data	Crop stock and dated, correctly-unitised market prices.	E2-S1 Manage my crop stock E2-S2 Government mandi price import and sync E2-S3 Price boards with units, dates and staleness
E3 Selling Decision: Net Return & Market Finder	Estimated net return per market, with assumptions shown.	E3-S1 Compare app markets by estimated net proceeds E3-S2 Market finder: origin, shortlist and road distance E3-S3 Market finder: cost breakdown, net and map
E4 Transport, Booking & Live Tracking	Book a truck, run the trip, track it live to delivery.	E4-S1 Truck listing and availability E4-S2 Booking request, trip status and delivery record E4-S3 Live trip tracking with operator consent
E5 AI Assistant, Admin Insights & Production Readiness	Multilingual assistant, admin console, and a safe deployment.	E5-S1 Multilingual agriculture assistant with voice input E5-S2 Admin console: users, documents and system metrics E5-S3 Production security, testing and deployment readiness

E1 · Accounts, Roles & Access Control

E1-S1 Sign up and sign in with role-based access

Must · 3 pts

As a **farmer or truck operator**, *I want* to create an account with my email and password and sign in, *so that* I only see and change the records that belong to my role.

AC1. Public sign-up offers only farmer and truck operator roles; other roles are set by an admin.
AC2. 5 failed sign-ins lock the account for 15 minutes.
AC3. The backend checks role and record owner on every protected request.

E1-S2 Admin OTP sign-in and password recovery

Must · 3 pts

As a **administrator or any user who forgot a password**, *I want* a one-time code sent to my email, *so that* admin access is protected and I can recover my account without help.

AC1. Admin sign-in needs the password and an emailed one-time code.
AC2. Codes expire in 10 minutes, allow 5 attempts and can be resent after 60 seconds.
AC3. A user can reset a forgotten password with an emailed code.

E1-S3 Verified truck-operator profile and documents

Must · 3 pts

As a **truck operator**, *I want* to complete my profile and upload my vehicle permit and both sides of my driving licence, *so that* farmers and admins can trust who is carrying the crop.

AC1. Profile fields, vehicle permit and both licence sides are mandatory; a missing field is rejected (422).
AC2. Images must be JPEG, PNG or WebP, up to 5 MB.
AC3. Documents can be replaced but not deleted; admins can review them.

E2 · Crop Stock & Market Price Data

E2-S1 Manage my crop stock

Must · 2 pts

As a **farmer**, *I want* to add, edit and delete my crops with quantity, pickup location and production cost per kg, *so that* the comparison, market finder and bookings start from what I actually have.

AC1. Farmers add, edit and delete only their own crops.
AC2. Each crop has quantity, pickup location and optional production cost per kg.
AC3. Profit is shown only when production cost is entered.

E2-S2 Government mandi price import and sync

Must · 5 pts

As a **administrator**, *I want* government mandi prices loaded from the dataset or the data.gov.in Agmarknet feed on a schedule, *so that* the price board and market finder use recent, de-duplicated published prices.

AC1. Prices import from the CSV dataset or the data.gov.in feed.
AC2. Duplicate rows are merged; records older than 30 days are removed.
AC3. Each run is logged; an admin can see status and start a sync.

E2-S3 Price boards with units, dates and staleness

Must · 3 pts

As a **farmer**, *I want* to browse government mandi prices and app-recorded market prices with their dates, units and trends, *so that* I can see what each market is paying and how old that price is.

AC1. Filter by state, district, market and commodity; view price history.
AC2. Government prices show in ₹/quintal, app prices in ₹/kg; the two are never mixed.
AC3. Every price shows its source and date; app prices older than 48 hours are flagged stale.

E3 · Selling Decision: Net Return & Market Finder

E3-S1 Compare app markets by estimated net proceeds

Must · 3 pts

As a **farmer**, *I want* to compare app-recorded markets by the estimated money left after selling costs, *so that* the highest price is not mistaken for the best market.

AC1. Net = price minus transport, loading, labour, commission, packaging and loss.
AC2. Every cost assumption used is listed next to the result.
AC3. The result is labelled an estimate, not a guaranteed price.

E3-S2 Market finder: origin, shortlist and road distance

Must · 5 pts

As a **farmer**, *I want* to give my location or a place name and get nearby mandis shortlisted by distance and price, *so that* I only compare markets I can realistically reach.

AC1. Origin comes from location, place name, crop pickup or profile (India only).
AC2. Markets are shortlisted by distance and price, then road distance is fetched.
AC3. If a market location is approximate, the result says so.

E3-S3 Market finder: cost breakdown, net and map

Must · 3 pts

As a **farmer**, *I want* each shortlisted market's price, transport cost and estimated net on a list and a map, *so that* I can see why one market leaves more money than another.

AC1. Each market shows price, distance, cost breakdown and estimated net.
AC2. Markets appear on a map; the nearest three show on the farmer dashboard.
AC3. If routing is unavailable, the app says so instead of showing old results.

E4 · Transport, Booking & Live Tracking

E4-S1 Truck listing and availability

Must · 2 pts

As a **truck operator**, *I want* to register my trucks with type, fares, status and current location, *so that* farmers can find a truck that fits their load.

AC1. Vehicle number, type, status, fares and location are required; no default fares.
AC2. Operators manage only their own trucks.
AC3. Farmers can see available trucks and their capacity.

E4-S2 Booking request, trip status and delivery record

Must · 5 pts

As a **farmer**, *I want* to book a truck for my crop to a chosen market and follow the booking until delivery, *so that* transport is arranged and the sale is recorded in my reports.

AC1. A farmer books a truck for a crop, quantity and market.
AC2. The operator accepts or rejects, then moves the trip: picked up → in transit → delivered.
AC3. Each status change sends a notification; deliveries feed the revenue chart.

E4-S3 Live trip tracking with operator consent

Should · 5 pts

As a **farmer waiting at the market**, *I want* to see where the truck is while it carries my crop, *so that* I know when it will arrive, and the operator shares location only when they agree to.

AC1. Location is shared only after operator consent, and only during the trip.
AC2. A GPS update is sent every 15 seconds; the farmer sees 'updated N s ago'.
AC3. The farmer is warned after 45 seconds with no update; delivery stops sharing.

E5 · AI Assistant, Admin Insights & Production Readiness

E5-S1 Multilingual agriculture assistant with voice input

Should · 5 pts

As a **farmer**, *I want* to ask farming questions by text or voice in my own language, *so that* I get short, relevant answers without the assistant inventing prices or schemes.

AC1. Answers stay on agriculture and never invent prices or schemes.
AC2. Voice questions are transcribed; replies come in the chosen language (10 languages).
AC3. If the main model is slow, a fallback model answers.

E5-S2 Admin console: users, documents and system metrics

Must · 3 pts

As a **administrator**, *I want* one console to manage users, review operator documents, watch usage and check mandi sync, *so that* I can keep the platform trustworthy without touching the database.

AC1. Shows user counts, assistant use and API requests for 24 h, 7 days or 30 days.
AC2. Admins change roles or remove users, but not themselves or the last admin.
AC3. Admins review operator documents and mandi sync status.

E5-S3 Production security, testing and deployment readiness

Must · 5 pts

As a **developer deploying the app**, *I want* production settings, security controls and automated checks in place, *so that* the live app is safe and a change cannot quietly break it.

AC1. Production mode hides API docs and rejects a weak secret key.
AC2. Only listed frontend origins are allowed; no secrets in the repo.
AC3. Gap: automated tests and versioned database migrations still to add.